Radiology

Introduction to Magnetic Resonance Imaging (MRI)

Magnetic Resonance Imaging (MRI), also referred to as nuclear magnetic resonance imaging (NMRI) or magnetic resonance tomography (MRT), is a highly sensitive and specific diagnostic modality. It is used to visualize the detailed internal structures of the body. Unlike computed tomography (CT) scans or conventional X-rays, MRI does not rely on ionizing radiation. Instead, it generates images of atomic nuclei inside the body using a powerful magnet and radio frequency (RF) waves.

There are four primary types of MRI machines:

- Closed MRI
- Open MRI
- Extremity MRI
- Dynamic MRI

How MRI Works

To create an MR image, the patient is placed inside a large scanner featuring a powerful permanent magnet. The most commonly used magnet operates at 1.5 Tesla, which is approximately 30,000 times stronger than the Earth's magnetic field.

- This strong magnetic field causes the nuclei of many atoms in the body, primarily hydrogen, to align themselves with the field.
- An RF pulse is then emitted, prompting some hydrogen nuclei to absorb the energy and resonate.
- Once the RF pulse is turned off, the absorbed energy is released from the body as a signal, which is detected by coils within the scanner.

Because the magnet is so powerful, all metallic objects, such as jewelry and clothing with metal zippers, must be removed before entering the MRI room.

Magnetic Signals and Tissue Contrast

The strength of the MR signal depends on the overall concentration of hydrogen nuclei (proton density) and how tightly bound the hydrogen is within the tissue.

- **Tightly bound hydrogen:** Found in bone, these atoms do not align well with the magnetic field and produce a weak signal.
- **Loosely bound hydrogen:** Found in liquids and soft tissues, these mobile atoms react strongly to the RF pulse, producing an intense signal and a brighter image.

Tissue contrast relies heavily on **T1** and **T2** relaxation times:

- **T1-Weighted Images:** These are mostly used to demonstrate anatomy. Tissues with short T1 times (like fat) appear bright, while tissues with long T1 times (like water, fluid, or CSF) appear dark.
- **T2-Weighted Images:** These are primarily used to identify pathology, such as inflammation, because pathological tissues generally contain more water. Tissues with long T2 times (like TMJ fluid or CSF) appear bright, whereas tissues with short T2 times (like fat) appear dark.

MRI Contrast Agents

To further enhance the appearance of inflammation, tumors, or blood vessels, contrast agents—most commonly Gadolinium—can be injected intravenously. They can also be injected directly into joints for arthrograms. Gadolinium is not imaged directly; rather, it shortens the T1 relaxation times of the surrounding tissues, causing them to appear brighter.

Comparing MRI and CT

Feature	**MRI**	**CT Scan**
Technology	Uses non-ionizing radio frequency (RF) signals.	Uses ionizing X-ray radiation.
Best Used For	Soft tissues and tumor detection/identification in the brain.	Tissue with a higher atomic number (bones, calcifications) and solid tumors in the chest or abdomen.
Contrast Agents	Uses paramagnetic properties (gadolinium, manganese) to alter tissue relaxation times.	Uses elements with a high atomic number (iodine, barium) for X-ray attenuation.
Logistics	Slower and more expensive.	Faster, less expensive, and more widely available. Suffers less from motion artifacts.

Advantages and Disadvantages of MRI

Advantages:

- It provides the best contrast resolution for soft tissues.
- It does not involve any harmful ionizing radiation.
- It allows for direct multi-planar imaging without needing to reposition the patient.

Disadvantages & Safety Concerns:

- Imaging times can be long, sometimes reaching up to 30 minutes.
- The procedure carries a high imaging cost.
- The narrow space and acoustic noise can cause claustrophobia or discomfort. This can be managed using open MRIs, sedation, anesthesia, or headphones playing music.
- Ferromagnetic metals inside the body (e.g., pacemakers, insulin pumps, cerebral aneurysm clips, shrapnel) are major hazards. The magnetic field can move these objects, cause excessive heating, or induce strong electrical currents. Note: Gold, stainless steel, and iron oxide are ferromagnetic, while titanium and amalgam are not.
- Tattoos containing iron oxide (found in some red, black, brown, or flesh-colored dyes) can cause a burning sensation because the magnetic metals can convert RF pulses into electricity or be "pulled" by the magnet.
- Gadolinium contrast agents should be avoided during pregnancy because they cross the placenta and enter the fetal bloodstream.

Clinical Applications

Due to its excellent soft tissue resolution, MRI is highly effective for evaluating:

- The integrity and position of the disk in the temporomandibular joint (TMJ).
- Soft tissue diseases, particularly neoplasia of the neck, cheek, tongue, and salivary glands.
- Malignant involvement in lymph nodes.
- Suspected early osteomyelitis of the jaws to detect edematous changes in fatty marrow and surrounding soft tissue.
- Blood flow in arteries (via MR angiography) to diagnose aneurysms, occlusions, or malformations